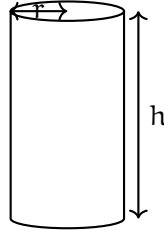


We consider the three steps of the modeling process.

Decision variables The design of the cylinder depends on two variables, both expressed in centimeters:

- the radius of the basis: r ,
- the height of the cylinder: h .



Objective function Since the thickness of the aluminium is the same at any part of the can, the total surface of the cylinder has to be minimized. The objective function must provide the total surface as a function of the decision variables:

$$f : \mathbb{R}^2 \rightarrow \mathbb{R} : f(r, h).$$

- Each basis is a circle of radius r , so its surface is πr^2 .
- The side of the can is a rectangle of area $2\pi r h$.

Therefore, the objective function to minimize is

$$f(r, h) = 2\pi r^2 + 2\pi r h.$$

Constraints The volume of the can must be 0.33 liters, that is 330 cm^3 . The first constraint is therefore:

$$\pi r^2 h = 330.$$

We also need non negativity constraints:

$$\begin{aligned} r &\geq 0, \\ h &\geq 0. \end{aligned}$$

The optimization problem is therefore:

$$\min_{r, h} 2\pi r^2 + 2\pi r h$$

subject to

$$\begin{aligned} \pi r^2 h &= 330, \\ r &\geq 0, \\ h &\geq 0. \end{aligned}$$

The optimal solution of this problem is $r = 3.746 \text{ cm}$ and $h = 7.491 \text{ cm}$.